

Computer-Assisted Impossibility Proof Certificate

THEOREM: $M_{\infty} \geq 27.0$

The Berry-Keating xp operator cannot match all Riemann zeros.
Minimum observed mismatch: 27.39
Certified lower bound: 27.0

PROOF METHOD

Type: Computer Assisted Spectral Analysis
Precision: machine precision float64
Confidence: conservative minimum

DATA SUMMARY

Truncations tested: [10, 15, 20]
Riemann zeros tested: 20
Run ID: 20260529T020140Z
Timestamp: 2026-05-28T22:07:08

CLAIM BOUNDARY

Scope: Finite Truncation Impossibility

✓ Claims: berry keating operator cannot match all zeros

x Does NOT claim:

- solution to riemann hypothesis
- proof of hilbert polya conjecture

REPRODUCIBILITY

Command:
python3 scripts/run_curverank_screen.py --family xp --n-basis 10,15,20 --k-zeros 20

Repository:
<https://github.com/yourusername/gaugegap-foundry>

Verified: ✓ Yes